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Question 1

- a) Languages which are similar to natural language statements, we called as High level language. Machine language and Assembly language are low level languages. Machine language does not need any translator, but Assembly language needs a translator. That translator we called as the assembler.

High Level Language	Low Level Language
A HLL is easily understandable by the human.	A Low Level Language is easily understandable by the computer.
High Level Language is a programmer friendly language.	Low Level Language is the language that is machine friendly.
HLL is not memory efficient.	Low Level Language is more memory efficient.
HLL requires a compiler or an interpreter to convert the program into machine code.	Assembly Language requires an assembler to convert the program to machine code while machine language is executed by the computer directly.

b) Variables :-

Variable mean unknown value. It is use for store value. We can't use variables as our wish. Variable has a name , a type, a size and a value. It is a visual representation.

Ex:- int x=10;

Pointer :-

It is a special type of variable. Always we can identify a pointer easily because before the variable name we use a '*'. It can store a memory address of an existing variable.

Ex:- int *p;

Code

```
#include<stdio.h>
//index no : 23781 irmdoilangasekara
int main()
{
    int x=10; // variable

    int *p; //pointer

    p=&x;
    printf("memory address is %p",p);
}
```

c) When we are coming to programming languages there are different generations. Computers are not familiar with human languages. If we write a program by using English statement, the processor can't understand it. It only understand binaries (0s and 1s). So we need to convert the source code into machine code. So we need a translator to do that process.

Ex : 1) Compiler

2) Interpreter

3) Assembler

```
d)    #include<stdio.h>
      //index no : 23781 irmdoilangasekara
      int main()
{
    // declare variables.....
    int x=10;
    char c='p';
    float f=4.99;
    double d=6.666;
    //output.....
    printf("%d\n",x);
    printf(" %c\n",p);
    printf("%f\n",f);
    printf("%lf\n",d);
}
```

Question 2

a)

```
#include<stdio.h>
```

```
//index no : 23781 irmdoilangasekara
```

```
int main()
```

```
{
```

```
    // Declare integer variables x, y, z and assign the value 10 in the y and 20 in the  
    z.
```

```
    int x=0 ,y, z;
```

```
    y=10;
```

```
    z=20;
```

```
    //Increment the variables x, y by one
```

```
    x++;
```

```
    y+=1;
```

```
    //assign the sum in z.
```

```
    z=x+y;
```

```
    //Assign the difference of x, y in z
```

```
    z=x-y;
```

```
    //increment the variables x, y by one
```

```
    x++;
```

```
    y++;
```

```
    //Increment the value x by one and assign the value in z.
```

```
    x+=1;
```

```
z=x;
```

```
// Display the values of x, y, and z
```

```
printf("%d %d %d",x,y,z) ;
```

```
}
```

b)

```
#include<stdio.h>
```

```
//index no : 23781 irmdoilangasekara
```

```
int main()
```

```
{
```

```
printf("Index number : 23781\n");
```

```
// declaring variables...
```

```
int count=1,num,positive=0,negetive,zero=0,sum=0;
```

```
float avg;
```

```
// process....
```

```
while (count<=20)
```

```
{
```

```
printf("enter %d number:",count);
```

```
scanf("%d",&num);
```

```
sum+=num;
```

```
if(num>0)
```

```
positive++;
```

```
else if (num<0)
```

```
negetive++;
```

```

        else
            zero++;

        count++;
    }
    avg=sum/20;

    printf(" The average is :%.2f\n Total positives:%d\n Total negatives:%d\n
Total zeros:%d\n ",avg,positive,negetive,zero);
    printf("Index Number : 23781");
}

```

Question 3

a)

WHILE

```

#include <stdio.h>
//index no : 23781 irmdoilangasekara
int main()
{
    int y=1;
    while(y <= 5){
        int x=1;
        while (x <= y){

```

```
        printf("*");
        x++;
    };
    y++;
    printf("\n");
}
}
```

DO WHILE

```
#include <stdio.h>

//index no : 23781 irmdoilangasekara

int main()
{
    int y=1;
    do{
        int x=1;
        do{
            printf("*");
            x++;
        }while (x <= y);
        y++;
        printf("\n");
    }while (y <= 5);
}
```


b)

```
#include<stdio.h>

//index no : 23781 irmdoilangasekara

int main()
{
    float no1,no2;
    char oper;
//input
    printf("Enter two numbers :\n");
    scanf("%f %f",&no1,&no2);
//do the choice
    printf("Select the operator: \n 1 Add\n 2 Subtract\n 3 Multiply\n 4 Divide\n
5 Exit \n");
    scanf(" %c",&oper);

    switch(oper)
    {
        case '1': printf("%f", no1+no2); break;
        case '2': printf("%f", no1-no2); break;
        case '3': printf("%f", no1*no2); break;
        case '4': printf("%f", no1/no2); break;
        case '5': printf("Ended"); break;

    }
}
```

Question 4

a)

```
#include <stdio.h>

//index no : 23781 irmdoilangasekara
int highest(int a,int b,int c) //variables
{
    if(a>b&& a>c)
        return a;
    else if (b>a&& b>c)
        return b;
    else
        return c;
}

int main()
{
    int n1,n2,n3;
    printf("23781\n"); // index number
    printf("Enter 3 numbers:"); // input three intergers
    scanf("%d %d %d",&n1,&n2,&n3);
    printf("Highest number:%d \n ",highest(n1,n2,n3));
    printf("237811\n"); // index number
}
```

```

b)    #include <stdio.h>

//index no : 23781 irmdoilangasekara
//create function
int high(int n1)
{
Printf("index number: 23781");
//find positive negative zero
if(n1>0)
printf("%d is a positive number.\n",n1);
else if(n1<0)
printf("%d is a negative number.\n",n1);
else
printf("%d is a zero.\n",n1);
}
int main()
{
//create array and variables
int num1[10],i;
//display index number
printf("index number = 23781\n");
//input values
for(i=0;i<10;i++)
{
printf("Enter number %d : ",i+1);
scanf("%d",&num1[i]);
}
//call function
for(i=0;i<10;i++)
{

```

```
high(num1[i]);  
}  
//display index number  
printf("\index no: 23781");  
}
```

Question 5

a)

```
#include<stdio.h>
```

```
//index no : 23781 irmdoilangasekara
```

```
int main()
```

```
{ //declaration of the variables and array
```

```
float prices[10],max=0,sum=0,low=0;
```

```
int i;
```

```
//input
```

```
for(i=0;i<10;i++)
```

```
{
```

```
    printf("enter the price of product %d\n",i+1);
```

```
    scanf("%f",&prices[i]);
```

```
    sum=sum+prices[i];
```

```
    max=low=prices[0];
```

```
    if(prices[i]>max)
```

```
        max=prices[i];
```

```
    else if(prices[i]<low)
```

```
        low=prices[i];
```

```
}
```

```
printf("prices of all items is %.2f\n",sum);
```

```
printf("the highest price is %.2f\n",max);
```

```
printf("the lowest price is %.2f\n",low);
```

```
}
```

b)

```
i. #include<stdio.h>
#define ROW 4
#define COL 4
//index no : 23781 irmdoilangasekara
int main()
{
    int arr[ROW][COL], i, j;

    for(i = 0; i < ROW; i++)
    {
        for(j = 0; j < COL; j++)
        {
            printf("Enter arr[%d][%d]: ", i, j);
            scanf("%d", &arr[i][j]);
        }
    }
    printf("\nEntered 2-D array is: \n\n");
    for(i = 0; i < ROW; i++)
    {
        for(j = 0; j < COL; j++)
        {
            printf("%3d ", arr[i][j] );
        }
        printf("\n");
    }
    // signal to operating system everything works fine

}
```

ii.

```
#include<stdio.h>
#define ROW 3
#define COL 4
//index no : 23781 irmdoilangasekara
int main()
{
    int arr[ROW][COL], i, j;

    for(i = 0; i < ROW; i++)
    {
        for(j = 0; j < COL; j++)
        {
            printf("Enter arr[%d][%d]: ", i, j);
            scanf("%d", &arr[i][j]);
        }
    }
    printf("\nEntered 2-D array is: \n\n");

    for(i = 0; i < ROW; i++)
    {
        for(j = 0; j < COL; j++)
        {
            printf("%3d ", arr[i][j] );
        }
        printf("\n");
    }
}
```